

Highlights

- *B. bassiana*, *B. pseudobassiana* and *A. muscarius* are pathogenic to *L. sericata*.
- *B. bassiana* is the most virulent isolate to blowfly.
- Carnauba wax increases *B. bassiana* and *A. muscarius* virulence against *L. sericata*.
- Conidia formulated with carnauba wax were effectively attached to blowfly cuticle.

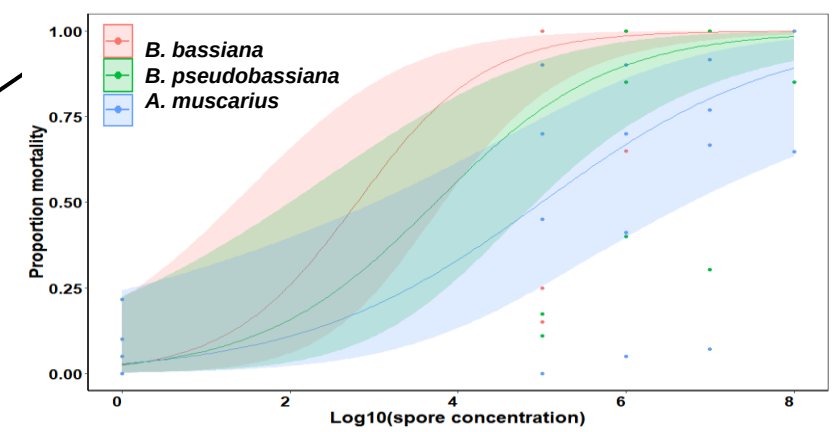
Dry conidia



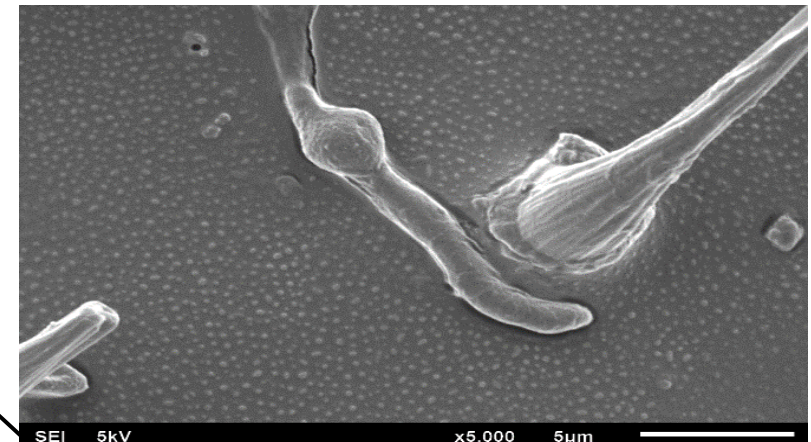
Carnauba wax

Treatment

Dead fly



Fly mortality



Scanning electron microscopy

Carnauba wax enhances the insecticidal activity of entomopathogenic fungi against the blowfly *Lucilia sericata* (Diptera: Calliphoridae)

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Abstract

Blowfly, *Lucilia sericata* (Diptera: Calliphoridae), is a problematic synanthropic insect pest, a vector of microbial pathogens, and the causal agent of secondary myiasis, which is a severe medical and veterinary injury. Fungal biopesticides are considered to be valid tools, alternative to synthetic pesticides, for the control of arthropod pests; however, to date, little is known about their bioactivity against blowflies. In this study, we assessed the insecticidal activity of three well-known entomopathogenic fungi, *Beauveria bassiana*, *Beauveria pseudobassiana* and *Akanthomyces muscarius* against *L. sericata*. In addition, we tested powdered carnauba wax as an electrically charged carrier to enhance the virulence of fungal spores. Pathogenicity tests on adult flies, by adult immersion in conidial suspension (10^8 conidia mL⁻¹), showed that the median lethal time (LT₅₀) was 5.3, 5.9, and 6.2 days for *B. bassiana*, *A. muscarius* and *B. pseudobassiana*, respectively. In topical tests, when 10^8 dry conidia were used with or without carnauba wax, the LT₅₀ was 7.7, 10.2, and 14 days without this carrier and 6.9, 8.6, and 13.8 days with it for *B. bassiana*, *B. pseudobassiana* and *A. muscarius*, respectively. Overall, our findings showed that, among the tested fungi, *B. bassiana* was the most virulent when formulated as a dry powder including carnauba wax, which greatly improved fungal efficacy against the blowfly. We discuss the utility of carnauba wax for electrostatic formulation powder of fungal spores in the integrated management of blowflies as an ecologically friendly tool to reduce the over-reliance on chemical insecticides.

Keywords: Electrostatic powder; *Beauveria bassiana*; *Akanthomyces muscarius*; *Beauveria pseudobassiana*; Biological control; Dust formulation.

1. Introduction

Myiasis is a severe medical and veterinary problem (Monteiro, 2017) caused by the presence of dipterous larvae in living vertebrates' tissues. This parasitism, that can be obligatory or facultative (Hope, 1840), causes pains and suffering to the hosts, serious damage to the tissues and, in more severe cases, permanent mutilation and septicemia (Hall et al., 2016). Human myiasis are mainly caused by fly species belonging to the Calliphoridae. In humans, myiasis are considered to be neglected diseases, since they are most often reported in patients living under low socioeconomic conditions or as a complication of neglected wounds (Hall et al., 2016; Singh and Singh, 2015). *Lucilia sericata* (Diptera: Calliphoridae) is a problematic synanthropic pest that causes myiasis. *Lucilia* larvae grow in necrotic tissues of neglected wounds and occasionally invade healthy tissues (Visciarelli et al., 2007). *L. sericata* represents the most important agent of sheep myiasis in northern Europe (Taylor et al., 2010). In sheep, *L. sericata* eggs are laid preferentially in the moist wool containing remains of urine and feces. After hatching, larvae develop into the wool causing fetid odor wounds that may attract other species of blowflies. In addition to its medical and veterinary importance, *L. sericata* is considered a study model for Calliphoridae flies, as it can be easily maintained under laboratory conditions (Silva et al., 2008).

Currently, the prevention of myiasis relies almost exclusively on synthetic insecticides (Sandeman et al., 2014). However, the inadequate use of these insecticides has increased insect resistance (Naqqash et al., 2016), environmental pollution and contamination of animal products (Desneux et al., 2007). The need for new biorational insecticides and the market-driven demand for organic products free of chemical residues reflect in the growing use of biocontrol agents. Entomopathogenic fungi may offer an environmentally sustainable alternative strategy for biological control of blowflies. Owing to the fact that few studies have addressed the control of calliphorid flies tied to the lack of available commercial mycoinsecticides to tackle this target around the world (Leathwick et al., 2019), there is an urgent need to seek microbial insecticides to aid in the sustainable management of this noxious insect.

One of the main challenges for the use of entomopathogenic fungi to control insect pests and vectors is the formulation and application of fungal propagules (Mascarin et al., 2019) to enhance both their virulence and persistence in the environment. In this sense, recent studies have shown that electrostatic powders can be effective in the formulating of the bioinsecticides (Barton et al., 2006; Meikle et al.,

2008), since insect cuticle accumulates electrical charges depending on the surface contact and flight time (McGonigle and Jackson, 2002). In this regard, carnauba wax has been shown as a potentially viable powder-based entomopathogenic fungi delivery system (Barton et al., 2006; Meikle et al., 2008; Athanassiou et al., 2017). In addition, carnauba wax powder has the advantage to be transferred from treated males to non-treated female of *Ceratitis capitata* (Diptera: Tephritidae) during mating behavior (Armstrong et al., 2008), which may improve horizontal transmission and dispersal of fungal spores.

The present study aimed to evaluate the insecticidal activity of three fungal pathogens – *Beauveria bassiana*, *Beauveria pseudobassiana* and *Akanthomyces muscarius* – against the myiasis-inducing blowfly *L. sericata*. In addition, we tested powdered carnauba wax as an electrically charged carrier to enhance the virulence of fungal spores as a novel formulation strategy to tackle this pest.

2. Materials and Methods

2.1 Insect rearing

L. sericata pupae were purchased from Koppert Italia S.R.L. (Verona, Italy). The pupae were held in cages until the emergence of the adults. The adults were provided with sugar and water *ad libitum* and kept under laboratory conditions (23 °C, 65 ± 5% RH, natural photoperiod) during the tests.

2.2 Culture and molecular identification of fungal isolates

Three isolates belonging to well-known entomopathogenic species were tested: *Beauveria bassiana* (ARSEF 9588) from the USDA-ARS Collection of Entomopathogenic Fungal Cultures (Ithaca, NY, USA), *Beauveria pseudobassiana* (DSM 6651) from the German Culture Collection (DSMZ, Braunschweig, Germany) and *Akanthomyces muscarius* (I515), these two last deposited in the Fungal Collection of the Plant Pathology & Mycology Lab of the Department of Agriculture, Food and Environment (DAFE) of the University of Pisa, Italy. All isolates were maintained on Potato Dextrose Agar (PDA) under mineral oil at 4 °C and cultivated on PDA supplemented with 1 g L⁻¹ yeast extract (PDAY) using Petri plates (90 × 15 mm) in the dark for 15 days at 25 ± 1 °C and RH ≥ 90 %.

Molecular identification of *B. bassiana* sensu stricto (s.str.) (ARSEF9588) and *A. muscarius* (I515) was performed by the internal transcribed spacers (ITS) sequencing. Genomic DNA from seven-day-old colonies grown on PDA at 25 ± 1 °C was extracted according to the Chelex 100 protocol, as described by Baroncelli et al. (2014). Complete internal transcribed spacers (ITS) 1 and 2 sequences, including the 5.8S gene of the nuclear ribosomal DNA were amplified and sequenced with primers ITS5 and ITS4 (White et al., 1990). Amplification and sequencing were performed according to Sarrocco et al. (2015). Sequences were deposited in GenBank with accession numbers MN960454 and MN960455 for *B. bassiana* s.str. (ARSEF 9588) and *A. muscarius* (I515), respectively. *B. pseudobassiana* DSM 6651 was already identified by ITS sequencing in a previous study (Bedini et al., 2018).

2.3 Pathogenicity test

Freshly produced conidia of each fungal isolate described in section 2.2 were harvested from fully sporulated colonies grown on PDAY, suspended in sterile 0.01% (v/v) Tween 80[®] aqueous solution and further filtered through a cheesecloth layer; 10 mL of the conidial suspension was adjusted to 10^5 , 10^6 , 10^7 or 10^8 conidia mL⁻¹. Twenty newly emerged unsexed adults of *L. sericata*, previously cold anesthetized (-4 °C for 3 minutes), were immersed for 10 seconds in 10 mL of each conidial suspension. Flies from control group were immersed in 10 mL of 0.01% (v/v) Tween 80[®] without conidia. Mortality was monitored and recorded daily for 10 days. The experiment was repeated four times on different days using new fungal cultures to ensure reproducibility.

2.4 Formulation with conidia and carnauba wax

Fresh conidia of *B. bassiana*, *B. pseudobassiana* and *A. muscarius* were harvested from PDAY as described above and suspended in 0.01 % (v/v) Tween 80[®]. Conidial suspensions of *B. bassiana* and *A. muscarius* were filtered through a cheesecloth layer and adjusted to the concentration of 10^8 or 10^9 conidia mL⁻¹. Since *B. pseudobassiana* produced fewer conidia than the other isolates, bioassays with *B. pseudobassiana* were performed by using only 10^8 conidia mL⁻¹ suspensions. One mL of each conidial suspension was poured into Petri dishes (30 × 10 mm) and put in a desiccator with activated silica gel for five days at 5 ± 0.5 °C. Carnauba wax powder (PVP Sociedade Anônima, Parnaíba, PI, Brazil) was freeze-dried at -40 °C for 24 h at

0.001 mbar. After drying, 10^8 and 10^9 conidia mL^{-1} of each isolate were mixed to 0.1 g of the freeze-dried carnauba wax; and this mixture yielded a final concentration of 10^9 and 10^{10} conidia g^{-1} , respectively. Water activity (a_w) of formulations was measured, and those with $a_w < 0.2$ were considered dried and able to maintain carnauba wax electrostatic.

Twenty newly emerged unsexed adults of *L. sericata*, previously cold anesthetized (-4°C for 3 min), were carefully transferred into a glass Petri dish containing the carnauba wax formulation with conidia or the dry conidia only. The flies were allowed to walk on the dry conidia or on the conidia-based carnauba wax formulation for 5 min. Two control groups were performed, one treated with 0.1 g of carnauba wax without conidia and one control group did not receive inoculum (no treatment). The flies were then removed from the plates and placed in small plastic cages (20 cm in diameter, 140 cm in height) which were closed, on the top, with a net cloth. Sugar and water were provided *ad libitum*. The flies were maintained at $24 \pm 1^\circ\text{C}$, with $\text{RH} \geq 90\%$ and natural photoperiod. Flies mortality was assessed daily for 15 days. The experiments were repeated three times in different days using new fungal cultures to ensure reproducibility.

2.5 Scanning electron microscopy (SEM)

To compare the potential sites of infection and the spatial distribution of dry or wax-formulated conidia on the integument surface of blowfly, *L. sericata* adults were treated with 10^8 dry conidia formulated with carnauba wax or only dry conidia, as described above. Flies were maintained at $24 \pm 1^\circ\text{C}$, with $\text{RH} \geq 90\%$ and natural photoperiod, for 48 h. After this incubation period, each fly was fixed by immersion in 2 mL of fixative [2 % (v/v) glutaraldehyde, 2 % (v/v) paraformaldehyde, 3 % (w/v) sucrose in 0.1 M sodium cacodylate buffer solution, pH 7.2] at 4°C for 10 days. Thereafter, samples were washed in phosphate buffer, dehydrated in graded series of ethanol solutions and subjected to critical point drying (Autosamdri[®]-815). Flies were metallized with gold in a sputter-applicator (Denton Vacuum Desk V) and analyzed by a scanning electron microscope (Jeol JSM 6610), located in the multiuser laboratory of high-resolution microscopy (LabMic/UFG, Goiânia-GO, Brasil), at the accelerating voltage of 20 kV.

2.6 Statistical analysis

Goodness-of-fit for model selection was assessed using half-normal plots with simulation envelopes (Moral et al., 2017) along with examination of the smallest Akaike's information criterion (AIC). All analyses were conducted in the free statistical environment R (Team, 2018). Probability of surviving was estimated using the censored survival data of *Lucilia* flies exposed to different fungal entomopathogens prepared at different concentrations as either aqueous conidial suspensions or as dry conidia. The dry conidia of these fungi were tested alone or combined with carnauba wax at two different concentrations. The survival curves and estimates of median and 90 % lethal times (LT₅₀ and LT₉₀) with their respective 95 % confidence intervals were obtained from Weibull parametric models using the “survival” package (Therneau, 2015). Survival curves were pairwise compared using log-rank test with *P*-values adjusted with Bonferroni-Holm method and considered significant at *P* < 0.05.

Mortality data recorded after 10 days of exposure were fitted to a generalized linear model with logit link function and a quasi-binomial distribution for errors to take into account the overdispersion parameter [i.e., variance greater than the mean in proportion data (Demétrio et al., 2014)]. In this model, the fixed effects were attributed to fungal species, concentration and their interaction term. Median and 90 % lethal concentrations (LC₅₀ and LC₉₀) values and their respective 95 % confidence intervals for fungal species were estimated using the function ‘dose.p’ from the “MASS” package (Venables and Ripley, 2002).

3. Results

3.1 Molecular identification of fungal isolates

Amplification of the ITS regions resulted in around 550 bp long sequences for both ARSEF9588 and I515 isolates. For isolate ARSEF 9588 the comparison of our ITS sequence (accession number MN960454) with those deposited in GenBank resulted in high similarity percentages (100 %) with ITS sequences from other isolates of *B. bassiana* (such as MK862360.1, MF802501.1, MF802497.1, MG515530.1, for instance). When ITS sequence of isolate I515 (accession number MN960455) was compared with those available in GenBank, a 100 % similarity percentage was achieved with ITS sequence of *A. muscarius* (accession number MH858370.1).

3.2 Pathogenicity to *Lucilia sericata* adults

Survival curves of flies inoculated with fungal spores differed significantly from the control group (Figure 1): A) *B. bassiana* ($\chi^2 = 289.36$, $df = 4$, $P < 0.001$), B) *B. pseudobassiana* ($\chi^2 = 214.8$, $df = 4$, $P < 0.001$) and C) *A. muscarius* ($\chi^2 = 131.12$, $df = 4$, $P = 0.002$). The sporulation rate of fly cadavers was higher than 90 % in all fungal treatments with no significant differences between fungal isolates.

Lethal rates of flies decreased with the increase in conidial concentration for all fungal isolates tested (Table 1). At 10^7 conidia mL^{-1} , the LT_{50} of flies treated with *B. bassiana* was 3 days lower than that of the flies treated with *A. muscarius* ($\text{LT}_{50} = 6.2$ and 9.1 days, respectively). Consistently, smallest value of LT_{90} was observed for the flies treated with *B. bassiana* ($\text{LT}_{90} = 7.9$ days at 10^8 conidia mL^{-1}), whereas, at the same conidial concentration, the LT_{90} 's of flies treated with *B. pseudobassiana* and *A. muscarius* were 9.0 and 10.2 days, respectively.

According to the logistic binomial model, fly mortality was significantly correlated with fungal conidial concentration with slopes significantly different ($\chi^2 = 50.02$, $df = 3$, $P < 0.001$) (Figure 2). In detail, *B. bassiana* had a significantly higher slope than *A. muscarius*, but not in relation to *B. pseudobassiana* (Table 2). *L. sericata* was indeed more susceptible to *B. bassiana* than to the other two fungi tested ($\chi^2 = 9.51$, $df = 3$, $P = 0.023$), according to LC_{50} and LC_{90} values (Table 2). As a result, *B. bassiana* required many fewer conidia to kill 50 % and 90 % of flies than *B. pseudobassiana* and *A. muscarius* (Table 2). On the contrary, it was not possible to determine the LC_{90} for *A. muscarius*, because fly mortality did not reach 90 %, thereby confirming that this fungus was not as virulent as the two *Beauveria* isolates.

3.3 Virulence of conidia formulated with carnauba wax carrier

Treatments with carnauba wax mixed with dry conidia and conidia alone significantly reduced survival of *L. sericata* adults compared to wax control without conidia ($\chi^2 = 295.29$, $df = 4$, $P < 0.001$) (Figure 3). Actually, carnauba wax alone caused very low mortality to the flies, 6.5 % (Figure 3); the control group (without treatment) reached the mortality of 15 %. In addition, regarding comparison between survival curves, the survival rate of insects treated with either *B. bassiana* or *A. muscarius* conidia mixed with carnauba wax was significantly lower than insects treated with fungal conidia alone when applied at 10^8 conidia ($P = 0.043$) and 10^9 conidia ($P < 0.0001$), respectively (Figure 3A, C). This result was not observed with flies treated with *B. pseudobassiana* conidia and carnauba wax formulation. In this last case,

survival rates of *L. sericata* adults were similar in both treatments, with or without carnauba wax ($P = 0.091$) (Figure 3B).

Estimated lethal times (LT₅₀ and LT₉₀) of flies treated with dry conidia alone were similar to those of flies treated with conidia-based carnauba wax formulation, with the only exception for *A. muscarius* formulated with carnauba wax that enhanced conidia activity by killing faster the flies when applied at 10⁹ conidia compared with dry conidia only (Table 3).

3.4 Scanning electron microscopy (SEM)

Conidia of the entomopathogenic fungi were found clustered and attached mainly to the thorax (Fig. 4A) and wings (Fig. 4B) of the treated flies. Conidia were completely absent from the eyes, head and legs of *L. sericata*. No evidence of conidial germination of *B. pseudobassiana* (Fig. 4C) and *A. muscarius* (Fig. 4D) after 48 h from inoculation was found. Conversely, germinated *B. bassiana* conidia (unformulated) were clearly observed presenting bipolar (bidirectional) germination and appressorial formation on the integument of flies (Fig. 4E and 4F).

Carnauba wax particles were irregular in shape and size; wax microaggregates were found on the wings (Fig. 5A) or attached to the thorax bristles (Fig. 5B) with or without conidia. Germinated conidia of *B. bassiana* (Fig. 5C and D) and *A. muscarius* (Fig. 5G and H) were observed when formulated with carnauba wax at 48 h post-application, but not for *B. pseudobassiana* (5E and F); this fungus apparently requires a longer time to resume germination on the insect cuticle.

4. Discussion

This is the first study that showed *B. bassiana*, *B. pseudobassiana* and *A. muscarius* conidial pathogenicity to *L. sericata* adults, but with different degrees of virulence. Although blowfly larvae correspond to the parasitic stage, sheep fleece is a complex and dynamic environment, including its own microbial flora and a range of skin-secreted compounds that may affect fungal viability (Jackson et al., 2010). Adults have been considered to be the most suitable target stage for the fungal biocontrol of blowfly (Sandeman et al., 2014). In our experiment, fly mortality rates reached 100% after 10 days of incubation after immersion in suspension of *B. bassiana* at 10⁷ conidia mL⁻¹. Similar control rates were obtained after treating *L. sericata* adults by immersion with *Metarhizium anisopliae* s.l. at 10⁷ conidia mL⁻¹ (Wright et al., 2004). Although the

satisfactory results obtained here, an immersion treatment of adult flies is not feasible for practical field uses; therefore, in our study we proposed a dry formulation of the fungal entomopathogens within the aim of a dispersal strategy to control calliphorid flies. Moreover, dry formulations of entomopathogenic fungi can survive longer during storage than in liquid preparations (Moore et al., 1995).

Beside the insecticidal activity of these entomopathogenic fungi, in this work we tested carnauba wax as an additive to enhance their effectiveness against blowfly. Our findings suggest that dry conidia of *B. bassiana* and *A. muscarius* exhibited an increased insecticidal performance against blowfly *L. sericata* when mixed with the powdered carnauba wax, but this virulence was dependent on the conidial concentration. The most pronounced improvement in the virulence using carnauba wax for conidial formulation was achieved with *A. muscarius*, which had been previously shown to be the least virulent in the first bioassays using immersion technique for contamination. These results indicate that carnauba wax powder is able to boost the insect infection rate of the fungal conidia, but conidial concentration and conidial physicochemical properties may affect the interactions with carnauba wax and fungal virulence. Therefore, additional studies are necessary to unravel these complex interactions between fungal spores and carnauba wax on a target host's cuticle.

Improvement in performance of the entomopathogenic fungi may be due to the hydrophobic and lipophilic properties of carnauba wax that could have increased the adhesion of conidia onto the hydrophobic cuticle of insect hosts (David-Henriet et al., 1998; Barreto et al., 2016). In line with our results, Athanassiou et al. (2017) also showed a high efficacy of formulation of *B. bassiana* conidia plus Entostat[®] and kaolin against stored-product beetles; this dry powder formulation induced mortality rates comparable to those of the chemical insecticide deltamethrin in field conditions. As suggested by Athanassiou et al. (2017), dry conidia formulated with carnauba wax may favor the protection of conidia against detrimental abiotic factors such as ultraviolet radiation and high temperatures, but we have not tested whether such ecological protection was afforded by carnauba wax on the fungal species studied here. In addition, *B. bassiana* mixed with carnauba wax was able to control the bee mite, *Varroa* sp., without affecting bee colony health (Meikle et al., 2008).

Carnauba wax may also act as an electrostatic carrier for delivery of infective fungal propagules, such as aerial conidia (Barton et al., 2006). In fact, insects can accumulate electrical charges on the surface of their body during flight by contacting a

dielectric surface, transferring ions via dermal glands and attaching charged particles (McGonigle and Jackson, 2002). Because of that, carnauba wax carrying hydrophobic aerial conidia of entomopathogenic fungi may facilitate adhesion and thus better dispersion of these propagules to initiate infection. However, based on our SEM results, we could not infer whether or not the wax enhanced the rate of conidial germination, but we observed that carnauba wax did not influence the deposition rate of conidia onto the insect surface, showing similar numbers of conidia found per insect body when either formulated or not with this carrier.

Another feature of entomopathogenic fungi is the ability of spreading among the insect population to healthy individuals that have not directly been in contact with the insecticide formulation, thus enhancing the effectiveness of the biological treatment (Bedini et al., 2018). In this regard, the dispersion of the fungal conidia through a powder formulation may facilitate such individual-individual pathogen transmission, or also known as horizontal transmission, which is an important attribute to fungal epizootics under field conditions. Armsworth et al. (2008) demonstrated that *C. capitata*, which visited baits with Entostat[®] (a carnauba-based product), took powder up on their bodies and also those flies that had not come into direct contact with electrostatic powder were contaminated through social interactions with contaminated flies or contaminated surfaces. Actually, within the first 24 h from treatment, we observed an increase of self-cleaning behavior (auto-grooming) in the flies treated with conidia plus carnauba wax formulation; the absence of conidia, confirmed by SEM investigation, indicate that the flies had routinely cleared their eyes, head and legs during wax-induced grooming behaviors.

It is conventionally well-known that the main route of infection of entomopathogenic fungi is through the insect cuticle where conidia - after adhesion to the cuticle by nonspecific hydrophobic and electrostatic interaction - germinate and penetrate the host (Mannino et al., 2019). In our study, SEM observations showed that conidia of *B. bassiana*, unlike the other two fungal species tested, germinated from opposite sides on the insect cuticle. Such bidirectional germination by *B. bassiana* conidia is documented in the literature; Talaei-Hassanloui et al. (2007) correlated this type of conidial germination with the low virulence of *B. bassiana* isolates because no evidences of fungal penetration were detected. In our study, however, *B. bassiana* infected and had a better performance against the blowfly than the other two fungal isolates tested here. Lastly, it is also important to note that *A. muscarius* formulated with

carnauba wax exhibited an improved virulence in terms of shortening survival time to kill 50% and 90% of fly population at a higher concentration of conidia, but carnauba was not able to significantly reduce the LT₅₀ and LT₉₀ for *B. bassiana* and *B. pseudobassiana*. Such incongruence among these three fungi interacting with carnauba wax should be studied in greater detail to elucidate how this powder formulation might further improve fungal efficacy against this and other fly pests of agricultural and veterinary importance using reduced fungal loads for application.

5. Conclusion

In summary, our findings highlight the virulence of *B. bassiana* against adults of *L. sericata* and confirm the ability of dust carnauba wax as a carrier that may enhance the insecticidal activity of conidia produced by the entomopathogenic fungi. These results may serve as a baseline to design strategies for controlling *L. sericata* by facilitating dispersal of entomopathogenic fungi conidia formulated with carnauba wax powder.

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9. Caption for figures

Fig. 1 Survival curves and mycosed flies due to infection by three entomopathogenic fungi isolates: (A) *Beauveria bassiana* ARSEF9588, (B) *B. pseudobassiana* DSM6651, (C) *Akanthomyces muscarius* I515. Conidial suspensions were prepared in 0.01 % (v/v) Tween 80[®]. Survival curves were pairwise compared using log-rank test with *P*-values adjusted with Bonferroni-Holm method and considered significant at $P < 0.05$.

Fig. 2 Mortality data recorded 10 days after exposure were fitted to a generalized linear model with quasibinomial distribution for errors to accommodate overdispersion, and the fixed effects were attributed to fungal species, concentration and their interaction term. Median and 90 % lethal concentrations (LC₅₀ and LC₉₀) values and their respective 95 % confidence intervals for fungal species were estimated using the function ‘dose.p’ from the MASS package in R.

Fig. 3 Survival curves of *Lucilia sericata* adults treated with carnauba wax (control), dry conidia or dry conidia mixed with carnauba wax of different entomopathogenic fungi species: (A) *Beauveria bassiana*, (B) *Beauveria pseudobassiana* and (C) *Akanthomyces muscarius*. Survival curves were pairwise compared using log-rank test with *P*-values adjusted with Bonferroni-Holm method and considered significant at $P < 0.05$.

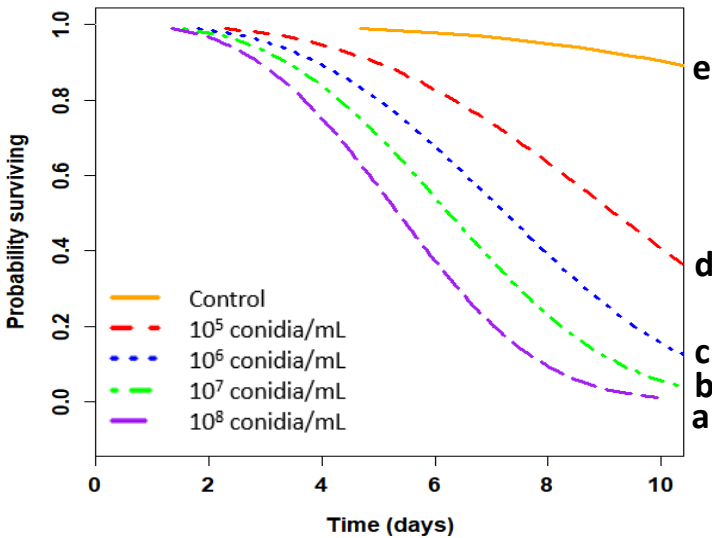
Fig. 4 Scanning electron microscopy of *Lucilia sericata* adults treated with dry conidia of different species of entomopathogenic fungi and incubated at 24 ± 2 °C, with RH = 90 % and natural photophase for 48 h. *Beauveria bassiana* (A, E and F), *Beauveria pseudobassiana* (C) and *Akanthomyces muscarius* (B and D). Regions of fly: (A) thorax and (B - F) wings. Con = conidia (single and clumps), Ap = appressorium, and H = fungal hypha.

Fig. 5 Scanning electron microscopy of *Lucilia sericata* adults treated with formulation of dry conidia with carnauba wax of different species of entomopathogenic fungi and incubated at 24 ± 2 °C, with RH = 90% under natural photophase for 48 h. Carnauba wax granules without conidia on wings (A) and thorax (B), *Beauveria bassiana* (C and

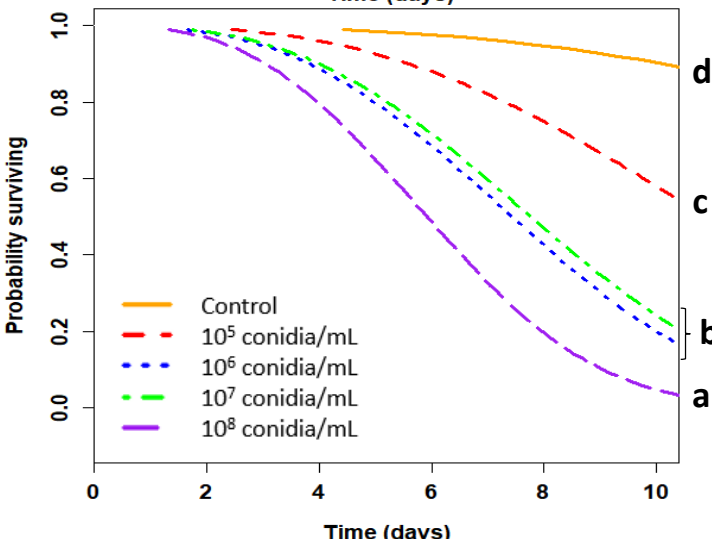
557 **D**), *Beauveria pseudobassiana* (**E** and **F**) and *Akanthomyces muscarius* (**G** and **H**). W =
558 carnauba wax granule; H = fungal hypha.

Figure 1

(A)



(B)



(C)

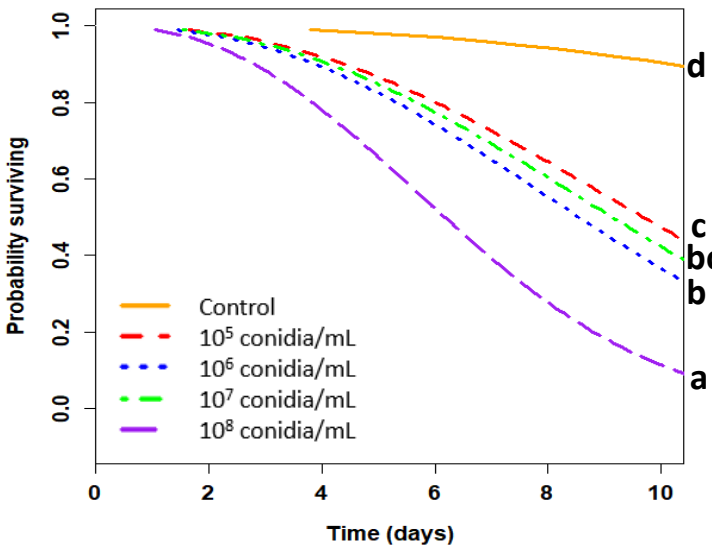


Figure 2

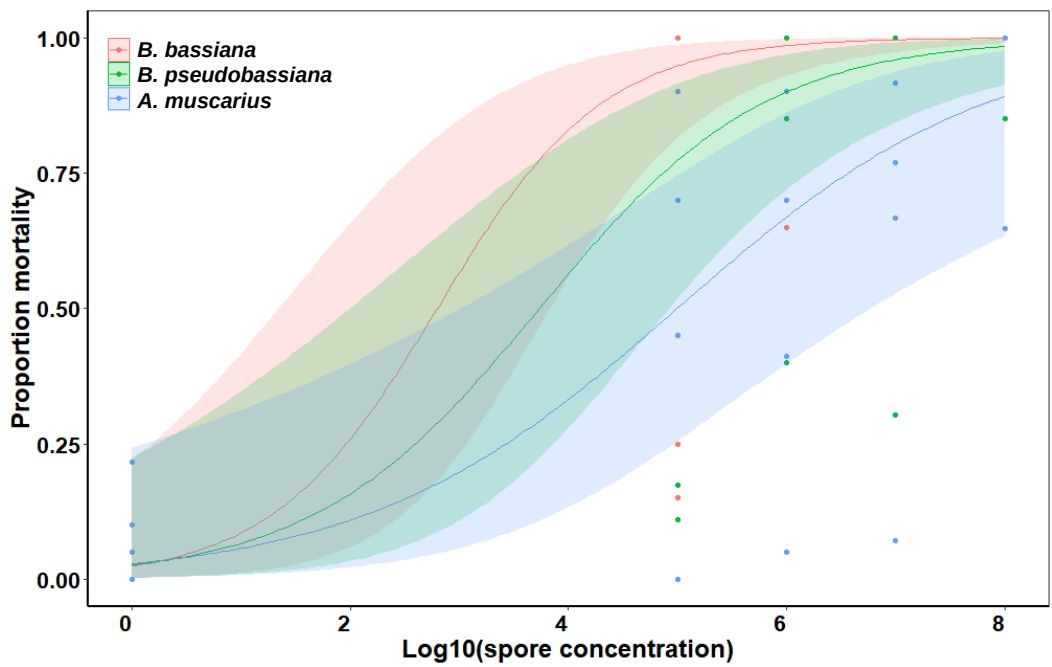


Figure 3

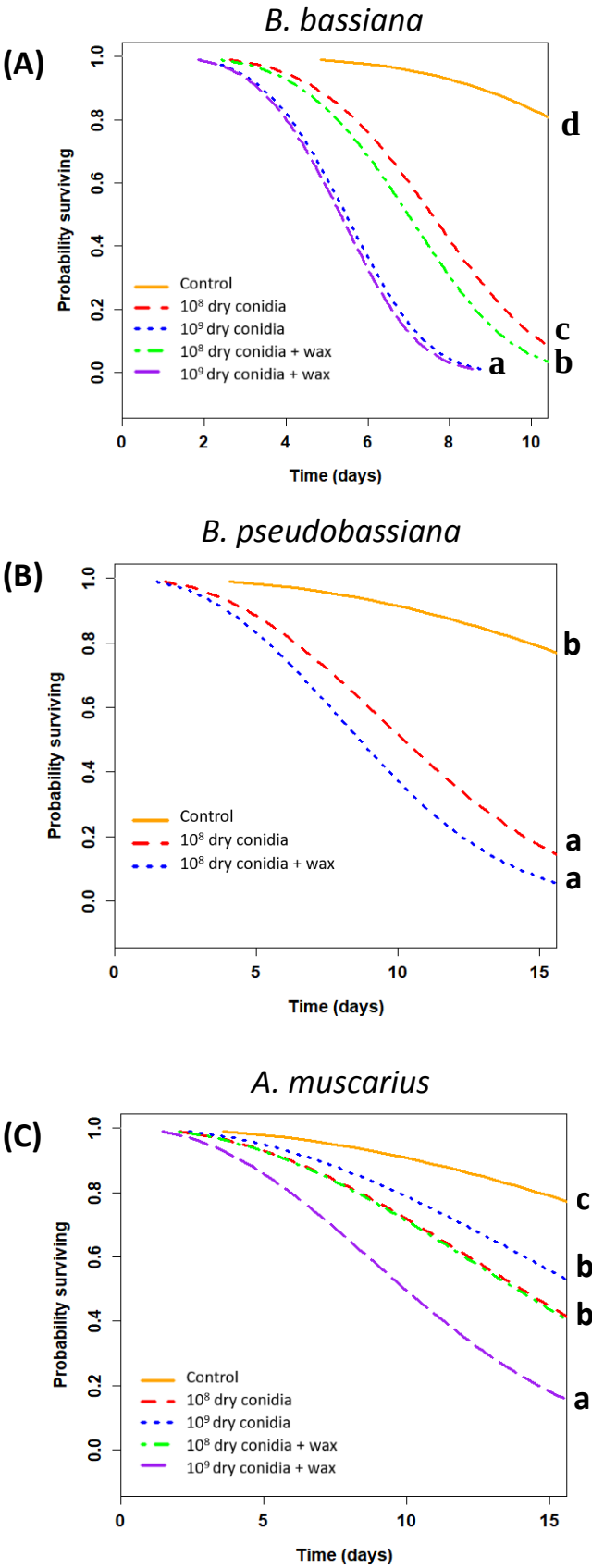


Figure 4

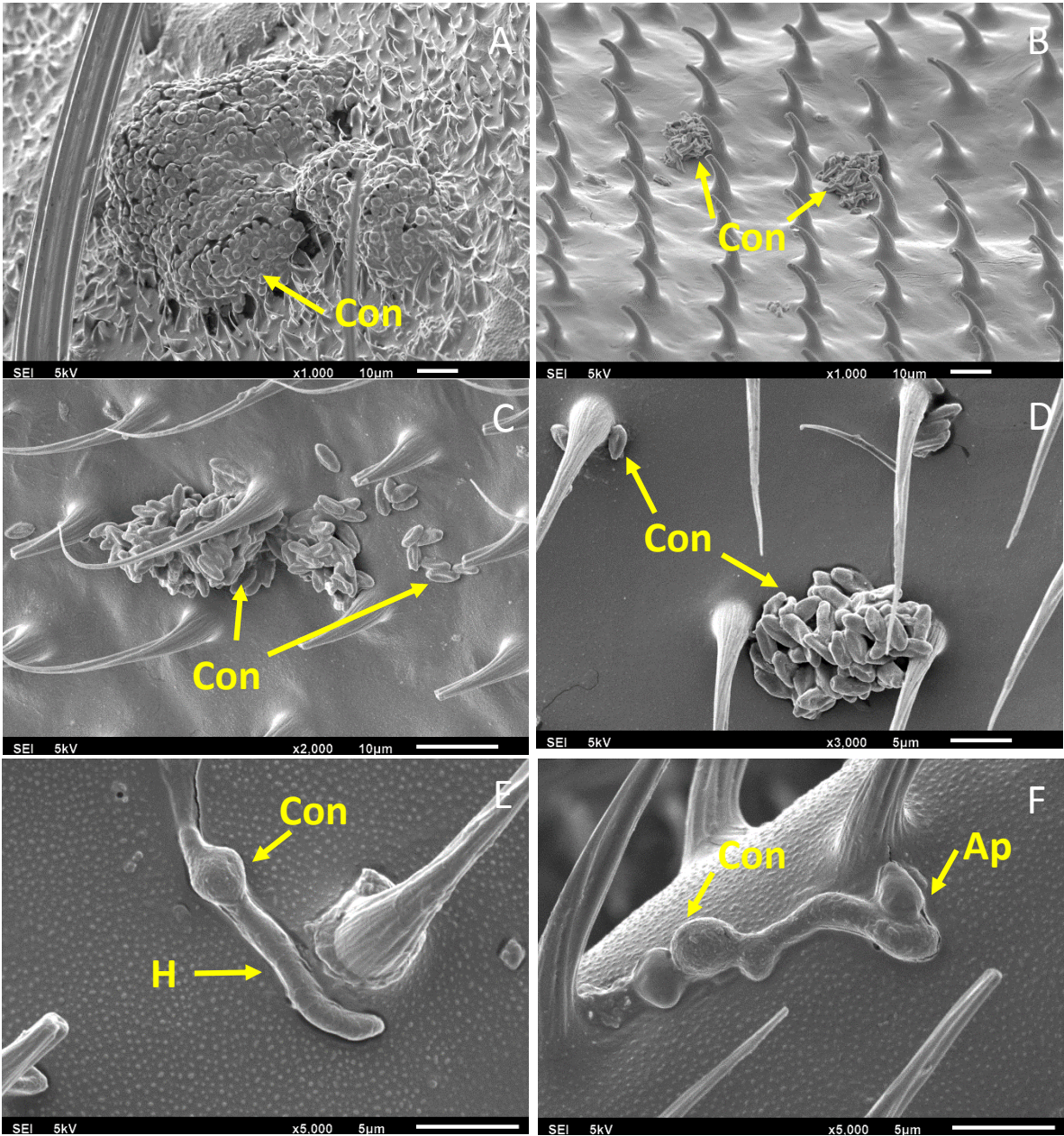


Figure 5

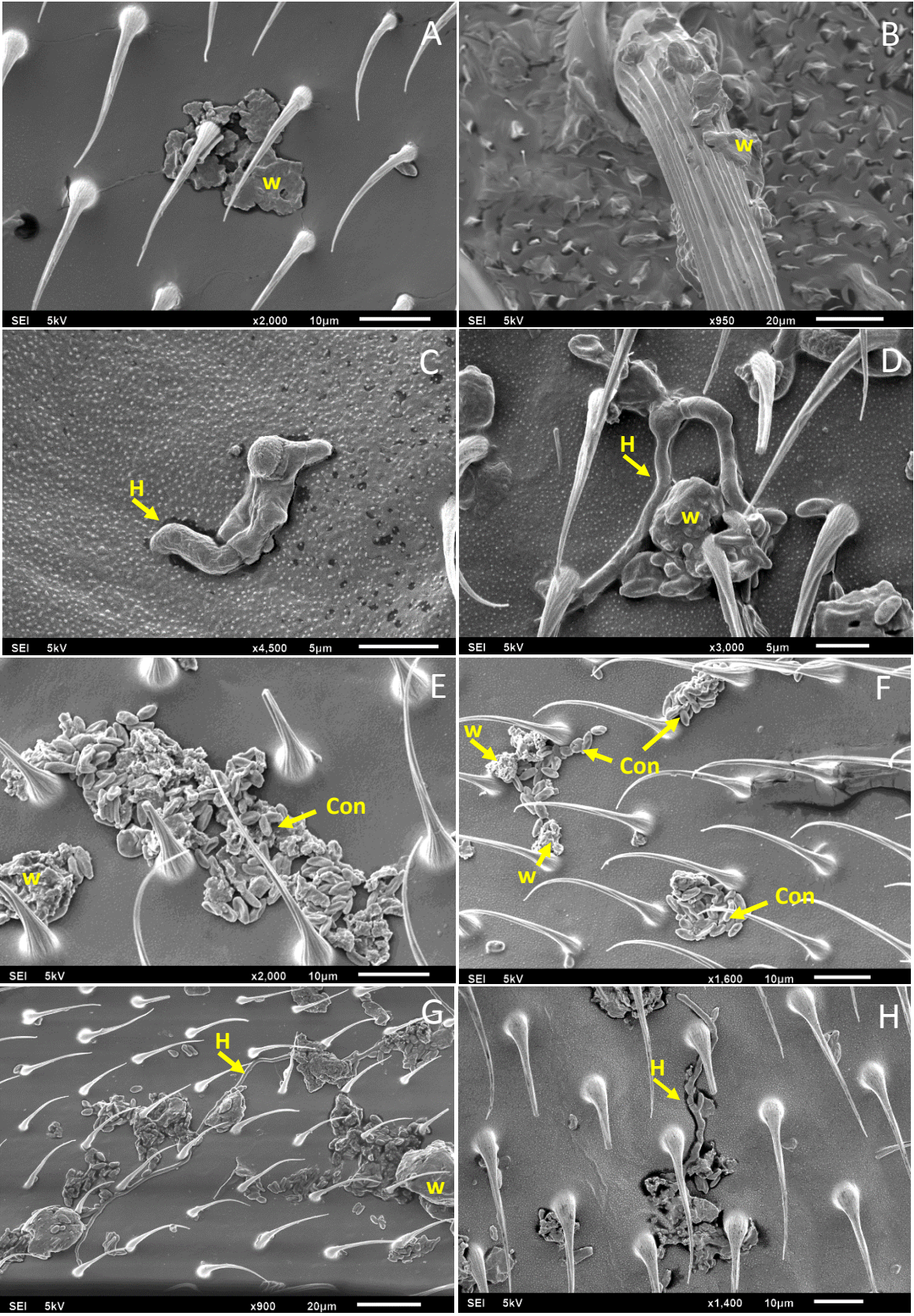


Table 1. Survival times (ST₅₀ and ST₉₀) and their respective confidence intervals (CI95%) of *Lucilia sericata* adults immersed in conidial suspensions at different concentrations (10⁵, 10⁶, 10⁷ or 10⁸ conidia mL⁻¹) incubated under laboratory conditions (24 ± 2 °C, 90 ± 5% RH, and natural photoperiod).

Fungus	Concentration (conidia mL⁻¹)	ST₅₀ (days)	CI95% (days)[†]	ST₉₀ (days)	CI95% (days)[†]
<i>B. bassiana</i>	0 (control)	ND*	-	ND*	-
	10 ⁵	9.2a	8.4 – 10.1	13.6a	12.3 – 15.0
	10 ⁶	7.2b	6.7 – 7.8	10.7b	9.9 – 11.6
	10 ⁷	6.2b	5.8 – 6.7	9.3b	8.6 – 10.0
	10 ⁸	5.3c	5.0 – 5.7	7.9c	7.4 – 8.5
<i>B. pseudobassiana</i>	0 (control)	ND*	-	ND*	-
	10 ⁵	10.9a	9.7 – 12.2	16.6a	14.7 – 18.9
	10 ⁶	7.4b	6.9 – 8.1	11.3b	10.4 – 12.4
	10 ⁷	7.8c	7.1 – 8.4	11.8b	10.9 – 12.9
	10 ⁸	5.9d	5.5 – 6.4	9.0c	8.4 – 9.7
<i>A. muscarius</i>	0 (control)	ND*	-	ND*	-
	10 ⁵	9.7a	8.5 – 11.0	16.0a	14.0 – 18.4
	10 ⁶	8.5a	7.6 – 9.6	14.2a	12.5 – 16.1
	10 ⁷	9.1a	8.1 – 10.3	15.1a	13.2 – 17.3
	10 ⁸	6.2b	5.6 – 6.8	10.2b	9.2 – 11.3

(*) ND (not determined) indicates that values could not be calculated because mortality was lower than 50% or 90% at the end of the experiment.

(†) Different letters within each fungal species indicate significant contrasts between spore concentrations, based on the overlap of 95% confidence intervals ($P < 0.05$).

Table 2. Virulence measured in median and 90% lethal concentration (LC₅₀ and LC₉₀) followed by confidence interval (CI95%) estimated for treatments of *Lucilia sericata* adults with each fungal species by immersion in aqueous conidial suspensions under laboratory conditions (24 ± 2 °C, 90 ± 5% RH, and natural photoperiod).

Fungus	Slope (± SE) ^a	LC ₅₀	CI95%	LC ₉₀	CI95%
		(conidia mL ⁻¹) ¹⁾		(conidia mL ⁻¹) ¹⁾	
<i>B. bassiana</i>	1.1 ± 0.2a	7.84 x 10 ²	1.42 x 10 ¹ – 4.15 x 10 ⁴	7.32 x 10 ⁴	7.22 x 10 ³ – 1.76 x 10 ⁵
<i>B. pseudobassiana</i>	0.82 ± 0.20ab	6.01 x 10 ³	6.02 x 10 ¹ – 6.06 x 10 ⁵	2.97 x 10 ⁶	1.74 x 10 ⁵ – 9.09 x 10 ⁶
<i>A. muscarius</i>	0.61 ± 0.19b	9.66 x 10 ⁴	4.21 x 10 ² – 2.26 x 10 ⁷	ND ^b	5.76 x 10 ⁶ - 3.74 x 10 ¹³

^a Slopes (± standard error) were obtained from the logistic-normal model and different letters indicate significant differences among treatments ($\chi^2 = 50.02$, df = 3, $P < 0.0001$). ^b ND = not determined, as mortality did not reach 90%

Table 3. Survival times (ST₅₀ and ST₉₀) and their respective confidence intervals (CI95%) of *Lucilia sericata* adults treated with dry conidia mixed or not mixed with carnauba wax and incubated under laboratory conditions (24 ± 2 °C, 90 ± 5% RH, and natural photoperiod).

Fungus	Concentration (number of conidia)	ST ₅₀ (days)	CI95% (days)	ST ₉₀ (days)	CI95% (days)
<i>B. bassiana</i>	0 (only wax)	ND	-	ND	-
	10 ⁸	7.7	7.1 – 8.1	10.2	9.6 – 10.9
	10 ⁸ + wax	6.9	6.5 – 7.4	9.4	8.9 – 10.0
	10 ⁹	5.5	5.1 – 5.8	7.4	7.0 – 7.8
	10 ⁹ + wax	5.3	5.0 – 5.7	7.2	6.8 – 7.6
<i>B. pseudobassiana</i>	0 (only wax)	ND	-	ND	-
	10 ⁸	10.2	9.0 – 11.5	16.8	14.8 – 19.1
	10 ⁸ + wax	8.6	7.7 – 9.7	14.2	12.7 – 16.0
<i>A. muscarius</i>	0 (only wax)	ND	-	ND	-
	10 ⁸	14.0	12.0 – 16.4	24.2	20.3 – 28.9
	10 ⁸ + wax	13.8	11.9 – 16.1	23.9	20.2 – 28.5
	10 ⁹	16.2	13.8 – 19.1	28.1	23.3 – 34.0
	10 ⁹ + wax	9.9*	8.6 – 11.5	17.2*	14.7 – 20.0

^a ND (not determined) indicates that values could not be calculated because mortality was lower than 50% or 90% at the end of the experiment. Asterisk (*) indicates significant difference based on the overlap of 95% confidence intervals ($P < 0.05$),